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Does Title Style Predict How Well a YouTube Video Performs? A Machine Learning Test

1. Inroduction

YouTube creators obsess over title style: capitalization, length, punctuation. The reason for this is believing it drives performance. I wanted to test this properly: does title style predict how well a video performs *relative to its channel's subscriber count*, in a way that is generalizable to almost all major channels across YouTube.

2. The Data

The dataset covers 17,933 videos across 93 channels, spanning 5 content categories (Gaming, Education, Documentary, Entertainment, News). I started with 25 channels and later added 68 more specifically to test whether findings held up on a broader, more diverse set rather than just describing those original 25.

The target metric is $\log(\text{views}) - \log(\text{subscribers})$, the goal of this is to normalize the results to account for the number of subscribers, a video from a 1M-subscriber channel could get 5x the views, and that might be seen as better than a 100M-subscriber channel getting 2x the views if the target metric used was just views/subscribers. Videos were grouped into 5 age buckets (<30d, 30-180d, 180-360d, 360-720d, 720d+) because a video's age heavily affects how many views it's had time to accumulate, and I didn't want that to go unaccounted for. I am also able to see trends throughout the different date ranges.

3. Initial Statistical Findings

The first pass was a straightforward regression: for each age bucket, fit an OLS model predicting the ratio from four title features: word count, average word length, percent of capitalized letters, and whether the title contains at least one ALL-CAPS word.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Fit on the original 25-channel sample, full sample, no train/test split.

From this data, I had hypothesized that having a shorter word count and shorter word lengths were going to be the factors that would have the most significant impact on getting more views. I also hypothesized that as we got past 6 months, all the affects start to be seen less and less and eventually become irrelevant, as the R^2 peaked at the 30-180 day range, and decreased from there.

Feature	<30d	30-180d	180-360d	360-720d	720d+
word_count	-0.476***	-1.163***	-0.450***	-0.534***	-0.100
avg_word_length	-0.257***	-0.556***	-0.315***	-0.124*	0.044
pct_caps	0.149*	0.066	-0.445***	0.194**	-0.004
has_allcaps_word	0.173**	-0.218*	-0.251**	0.043	-0.067
R ²	0.108	0.237	0.122	0.120	0.009
n (videos)	1015	1099	897	1137	848

4. The real test: does it generalize?

I ran channel-grouped 5-fold cross-validation on the full 93-channel dataset: each fold trains on ~74-75 channels and tests purely on the remaining ~18-19, so a channel never appears in both sets. A Ridge regression on the same 4 features was compared against a baseline that predicts the training-fold's average ratio for every video.

Age	n	Baseline		Improve-
Bucket	n Videos	Channels	R ²	ment
<30d	2,598	80	-0.099	+0.023
30-180d	4,492	82	-0.027	-0.036
180-360d	2,525	69	-0.185	-0.090
360-720d	2,842	59	-0.178	-0.015
720d+	5,150	51	-0.022	-0.026

The model beat the baseline in only 1 of 5 buckets, and even there the gain was marginal. In every other bucket, adding title features made predictions *worse* than just guessing the training mean. The negative R² values throughout (including for the baseline) reflect just how different held-out channels are from the channels used to train.

The apparent findings from Section 3 did not survive this test, there is no evidence to suggest title style significantly affects the view count of YouTube videos.

5. Channel Identity

To understand why, I decomposed the total variance in the ratio into two parts: the share explained just by knowing which channel a video came from (between-channel), and everything else, including title style (within-channel).

Age Bucket	n Videos	n Channels	% Variance Between- Channel
<30d	2,598	80	58.8%
30-180d	4,492	82	72.8%
180-360d	2,525	69	78.6%
360-720d	2,842	59	61.8%
720d+	5,150	51	46.2%

Channel Identity alone explains 46-79% of the variance in this ratio, depending on age bucket. That's the ceiling on what any per-video feature, title style included, could ever hope to explain for a video from a new channel: the majority of what makes a video's ratio high or low is a factor relative to the specific channel that posted it, not a general trend about title style. This also explains part of the earlier cross-validation results directly: the bucket with the *lowest* between-channel share (720d+, 46%) is exactly where title features had the most (if still limited) room to matter(highest Model R^2), while the bucket with the *highest* share (180-360d, 79%) showed the worst generalization(lowest Model R^2).

6. Conclusion

If you are interested in a specific small subset of YouTube channels, title style CAN be used as a statistically significant factor to predict video performance with a views-per-subscriber ratio, but these associations do not reliably generalize to channels outside that sample. True determinants for video performance using my target metric will be primarily be specific to each individual channel.

7. Limitations

- 93 channels is broad, but still a modest number of groups for channel-level claims, cross-validation folds held out only 18-19 channels each, which is a fairly small, noisy test set per fold. Further analysis could be done with a greater dataset.
- Category and channel composition aren't perfectly balanced. (News is underrepresented in the expanded sample)
- Only 4 title features were tested (word count, average word length, percent capitalized, presence of an all-caps word). Further analysis could have other title factors that potentially are statistically significant.